

MATH 2130 - Week 5

Positive Definite Matrices

1 Overview

We're going to return to what we did during the first couple of weeks here, returning to Gaussian Elimination and matrix factorizations, wrapping things up with the beautiful and powerful topic of **positive definite matrices**.

We've already seen that the matrix A can be factored into the product of a lower triangular, L , and upper triangular matrix, U , if there were no row exchanges needed: $A = LU$. When A is symmetric, this factorization is even nicer: $A = LDL^T$, where D is a diagonal matrix. We've seen how the entries of L, U , and D are related to Gaussian Elimination. Now we will take things one step further.

One thing that needs to be stressed here is that not needing to exchange rows corresponds to having no nonzero pivots. This was the key. After each row operation of adding a multiple of one row to the row below it left that lower row with a nonzero value in the pivot position. But, the pivots could be negative. Now we are going to look at what happens when all of the pivots are positive (which means strictly positive: $d_i > 0$ for all pivots d_i). When this happens, the matrix A is called **positive definite**.

Summarizing, the main points are the following: If A is a square matrix with no nonzero pivots, then

1. If the multipliers l_{ij} (the negatives of multiples of the pivot row used to eliminate the values below it) are put into a lower triangular matrix L , with ones on the main diagonal, the original matrix A is linked to the echelon form upper triangular matrix U by $A = LU$.
2. If A is **symmetric** and the diagonal matrix D containing the pivots is divided out of U , it leaves the transpose of L . Thus U is linked to L by $U = DL^T$ and we have the symmetric factorization $A = LDL^T$.
3. If the pivots are all positive (and A is symmetric still), then A is **positive definite**.

So far, we haven't given any indication of what special properties positive definite matrices have or why we should care about them. The answers to those questions are the main points of this week.

The Take-Home Exam is due on Wednesday, so please be sure to keep that in mind. On Thursday, we will spend some time wrapping the semester up and reviewing for the cumulative final exam, which will be on Friday. The Final Exam **WILL** include material covered this week in class, so please make sure that you take the time to understand what we are doing and ask any questions that you might have.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Positive Definite Matrices. Maxima, Minima, and Saddle Points.
- (T) Semidefinite and Indefinite Matrices.
- (W) Minimum Principles and Least Squares. **Take-Home Exam DUE.**
- (R) Wrap-Up and Review.
- (F) **Final Exam**

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Austin and Olver.
- ☐ Complete all of the homework exercises.

3 Key Concepts

- Positive Definite Matrices
- Quadratic Forms
- Completing the Square
- Rank One Matrices of the form ldl^T
- Matrix Multiplication (in terms of outer products)
- Minimum Principles
- Least Squares
- Normal Equations
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4 Important Definitions, Formulas, and Theorems

- A symmetric matrix A is **positive definite** if any of the following holds (they all imply the others):
 1. $x^T Ax > 0$ for all nonzero vectors x .
 2. All the eigenvalues of A satisfy $\lambda_i > 0$.
 3. All the principal submatrices A_k have positive determinants.
 4. All the pivots (without row exchanges) satisfy $d_i > 0$.
 5. There exists a nonsingular matrix W such that $A = W^T W$.
- Let A be an $n \times n$ matrix. The **principal submatrices** $A_k, 1 \leq k \leq n$ are the $k \times k$ matrices obtained from A by taking all entries in the first k rows and the first k columns of A . So, $A_1 = [a_{11}]$,

$$A_2 = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad A_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad \text{and so on, until } A_n = A$$

- If A is positive definite and its unit eigenvectors are the columns of the matrix Q . We have that $A = Q\Lambda Q^T$, with Λ the diagonal matrix of (positive) eigenvalues. Then the rotation $y = Q^T x$ produces the sum of squares

$$x^T A x = x^T Q \Lambda Q^T x = y^T \Lambda y = \lambda_1 y_1^2 + \cdots + \lambda_n y_n^2$$

- Every symmetric positive definite matrix A allows a symmetric factorization

$$A = LDL^T = l_1 d_1 l_1^t + \cdots + l_n d_n l_n^t$$

The numbers $d_i > 0$ are the pivots in D , and the vectors l_i are the columns of the matrix L .

5 Examples

- We'll start with the simplest possible case: A is a 2×2 symmetric matrix. In this case, we have

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

In order for Gaussian elimination to proceed without any row exchanges, we need $a \neq 0$. In this case, we can remove the b below the pivot by multiplying row 1 by b/a and subtracting (technically, we are multiplying by $-b/a$ and adding, but the signs work out nicer if we think of it as b/a and subtracting). This gives us

$$U = \begin{bmatrix} a & b \\ 0 & c - b^2/a \end{bmatrix}$$

for the echelon form (which is upper triangular) with pivots $d_1 = a, d_2 = c - b^2/a$. We also have that $l_{21} = b/a$. Putting into the lower triangular matrix L , we have

$$L = \begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix}$$

You can quickly check (do this!) that $A = LU$.

If we divide the pivots out of U we get

$$U = \begin{bmatrix} a & b \\ 0 & c - b^2/a \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & c - b^2/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix} = DL^T \quad (\text{check this!})$$

Putting it all together, we have

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ b/a & 1 \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & c - b^2/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix} = LDL^T \quad (\text{check this!})$$

By the property that $(AB)^T = B^T A^T$, we see that $(LDL^T)^T = L^{TT} D^T L^T = LDL^T$, we see that any matrix that factors as LDL^T must necessarily be symmetric (which we already knew about A).

So far, all we've done is show that $A = LDL^T$ when A is a 2×2 symmetric matrix with nonzero first pivot, $d_1 = a \neq 0$. We will now insist that both pivots are positive: $a > 0, c - b^2/a > 0$. We'll see

why we want to do this soon enough. Please note that it is not enough for just $a > 0$, $c > 0$. If c is “too small” and/or b is “too big,” then $c - b^2/a$ can be negative. Also note, if we rewrite the inequality $c - b^2/a > 0$ as $c > b^2/a$ and multiply through by a we get $ac > b^2$ which becomes $ac - b^2 > 0$. But for our matrix A we have that $\det A = ac - b^2$, so having a positive second pivot is equivalent to A having a positive determinant.

We’ll take a seeming detour here, looking at what are called **quadratic forms**. Let $f(x_1, x_2) = ax_1^2 + 2bx_1x_2 + cx_2^2$. Then it is easy to check that

$$f(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \text{or} \quad f = x^T A x \quad (\text{check this!})$$

- Okay, let’s look at some specific examples. Let

$$A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix}$$

We can check that this is positive definite by computing the determinants of the two principal submatrices:

$$\det A_1 = 2 > 0, \quad \det A_2 = \det A = 6 > 0$$

so A is positive definite.

Doing the normal Gaussian Elimination and factorization gives us

$$A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = LDL^T \quad (\text{check this!})$$

We also have that

$$x^T A x = 2(1x_1 + 2x_2)^2 + 3(0x_1 + 1x_2)^2 \quad (\text{check this!})$$

Please note the following:

- The coefficients outside the squared terms are exactly the values in the diagonal matrix, which are also the pivots of A .
- The coefficients in the first squared term (The $(1x_1 + 2x_2)^2$ term) are exactly the entries in the first column of L .
- The coefficients in the second squared term (The $(0x_1 + 1x_2)^2$ term) are exactly the entries in the second column of L .

This is amazing! The values in D and L give us everything that we need to know to find the quadratic form. This will always happen, no matter the size of A .

- Now let’s consider the 3×3 matrix

$$B = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 9 \end{bmatrix}$$

We will use a slightly different method to show that this matrix is positive definite. It involves a beautifully simple idea: the matrix B can be written as the sum of 3 matrices, each of which is a **rank one** matrix. The amazing thing is that each of these rank one matrices can be written in the form ldl^T , where l is a column vector and d is one of the pivots of B .

Let's see this in action. First, note that $d_1 = b_{11} = 1 > 0$. Take the first column of B and divide each entry by d_1 . This gives us, in general,

$$l_1 = \begin{bmatrix} 1 & a_{12}/d & \cdots & a_{1n}/d \end{bmatrix}$$

In the particular case that we are looking at, we have (since $d_1 = 1$)

$$l_1 = \begin{bmatrix} 1 & 2 & 2 \end{bmatrix}$$

Now form the matrix M_1 in the following way

$$M_1 = l_1 d_1 l_1^T = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} 1 \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$$

The matrix B can be written now as $B = M_1 + C$ and we can solve for C :

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 3 \\ 0 & 3 & 5 \end{bmatrix}$$

Note that C is symmetric and that the first row and columns are all 0's. Repeat the above process with the matrix C , taking $d_2 = c_{22} = 3 > 0$ and l_2 the second column of C divided by d_2 , to get M_2 . We have then that

$$M_2 = l_2 d_2 l_2^T = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} 3 \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 3 \\ 0 & 3 & 3 \end{bmatrix}$$

This means that $B = M_1 + M_2 + D$ and we solve for D to see that $D = M_3$ with

$$D = M_3 = l_3 d_3 l_3^T = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} 2 \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Putting this all together, we have

$$B = M_1 + M_2 + M_3 = l_1 d_1 l_1^T + l_2 d_2 l_2^T + l_3 d_3 l_3^T = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 3 \\ 0 & 3 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

There's a lot going on here. Let's take a look at everything that's happening. First, if we put the columns l_i together into a matrix we get

$$L = \begin{bmatrix} l_1 & l_2 & l_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix}$$

This is exactly the matrix L that shows up in the symmetric factorization $B = LDL^T$. What's more, if we put the values d_i into a diagonal matrix we get

$$D = \begin{bmatrix} 1 & & \\ & 3 & \\ & & 2 \end{bmatrix}$$

This is exactly the matrix D in the symmetric factorization.

Turning to the associated quadratic form, we have

$$f(x_1, x_2, x_3) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1^2 + 4x_1x_2 + 4x_1x_3 + 7x_2^2 + 14x_2x_3 + 9x_3^2$$

Looking at the quadratic forms associated to the M_i we have

$$f_1(x_1, x_2, x_3) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 1(x_1 + 2x_2 + 2x_3)^2 \quad (\text{check this!})$$

$$f_2(x_1, x_2, x_3) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 3 \\ 0 & 3 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 3(0x_1 + 1x_2 + 1x_3)^2 \quad (\text{check this!})$$

$$f_3(x_1, x_2, x_3) = \begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 2(0x_1 + 0x_2 + 1x_3)^2$$

Notice that the coefficients on the outside of the squared term are exactly the values of the associated d_i in the matrix D , and that the coefficients inside the squared term are exactly the entries in the associated column l_i from the matrix L .

Putting all of this together, we have that the quadratic form $f(x_1, x_2, x_3)$ can be written as the sum of squares $f_1(x_1, x_2, x_3) + f_2(x_1, x_2, x_3) + f_3(x_1, x_2, x_3)$:

$$x_1^2 + 4x_1x_2 + 4x_1x_3 + 7x_2^2 + 14x_2x_3 + 9x_3^2 = (x_1 + 2x_2 + 2x_3)^2 + 3(x_2 + x_3)^2 + 2x_3^2 \quad (\text{check this!})$$

Since the right-hand side is the sum of squares, it is **always** positive as long as $x = (x_1, x_2, x_3)^T \neq \vec{0}$. This means that the original matrix B is positive definite: $x^T B x > 0$ if $x \neq \vec{0}$.

- The previous example was long, but it is very instructive. If A is an $n \times n$ positive definite matrix, then A can always be written as the sum

$$A = M_1 + M_2 + \cdots + M_n$$

where each M_i is symmetric and the first $i - 1$ rows and columns of M_i are entirely zero. In fact, each M_i can be written as

$$M_i = l_i d_i l_i^T$$

with d_i the i th pivot of A and l_i the i th column of M_{i-1} divided by d_i (here, we let $M_0 = A$, the original matrix, so l_1 takes the first column of $M_0 = A$ and divides it by d_1 , the a_{11} entry of A).

- One last note. When we first defined matrix multiplication of two matrices A and B , we did it in terms of the inner product of the rows of A and the columns of B . If we let r_i be the i th row of A and c_j the j th column of B , then $(AB)_{ij} = \langle r_i, c_j \rangle$. So every entry of the product AB can be given in terms of the inner products of rows of A and columns of B . Finally, remember that

$$\langle x, y \rangle = x^T y$$

What we have just done above can be extended to show that the matrix product AB can also be written as a sum of **outer products** of the columns of A times the rows of B . Let $c_i = (a_{1i}, a_{2i}, \dots, a_{ni})^T$ now be the i th column of A and $r_j = (b_{j1}, b_{j2}, \dots, b_{jn})$ the j th row of B (please note that we are switching which matrix the rows and columns are coming from). Then the outer product of c_i and r_j is given by

$$c_i r_j^T = \begin{bmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{ni} \end{bmatrix} \begin{bmatrix} b_{j1} & b_{j2} & \cdots & b_{jn} \end{bmatrix} = \begin{bmatrix} a_{1i}b_{j1} & a_{1i}b_{j2} & \cdots & a_{1i}b_{jn} \\ a_{2i}b_{j1} & a_{2i}b_{j2} & \cdots & a_{2i}b_{jn} \\ \vdots & \vdots & \ddots & \vdots \\ a_{ni}b_{j1} & a_{ni}b_{j2} & \cdots & a_{ni}b_{jn} \end{bmatrix}$$

We can then express the product AB in the following way:

$$AB = c_1 r_1^T + c_2 r_2^T + \cdots + c_n r_n^T$$

You might want to check this for $n = 2$ and $n = 3$. Here's the case with $n = 2$

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}$$

The outer product of the first column of A and the first row of B gives

$$\begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} \\ a_{21}b_{11} & a_{21}b_{12} \end{bmatrix}$$

while the outer product of the second column of A and the second row of B gives

$$\begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} \begin{bmatrix} b_{21} & b_{22} \end{bmatrix} = \begin{bmatrix} a_{12}b_{21} & a_{12}b_{22} \\ a_{22}b_{21} & a_{22}b_{22} \end{bmatrix}$$

Adding these together gives

$$\begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} \\ a_{21}b_{11} & a_{21}b_{12} \end{bmatrix} + \begin{bmatrix} a_{12}b_{21} & a_{12}b_{22} \\ a_{22}b_{21} & a_{22}b_{22} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix} = AB$$

6 Sage Code

6.1 Commands

These are the most common Sage commands that we will use this week. Think of this more as a reference. It is usually easier to understand what the commands do by seeing them used in example code, which is given below.

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6.2 Example Code

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7 Additional Comments

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8 Reading Notes

- Olver 3.4 - Positive Definite Matrices
- Olver 5.1 - Minimization Problems
- Olver 5.2 - Minimization of Quadratic Polynomials
- Olver 5.3 - The Closest Point
- Olver 5.4 - Least Squares

9 Homework Exercises

Please do **ALL** exercises. Not all of them will be graded for correctness, but they will all be checked for completion. Failure to do all of them will lower the completion portion of the Homework score.

Please write your solutions up clearly and cleanly, using full English sentences. Try not to have a bunch of arrows pointing from one thing to another or a bunch of things crossed-out and a bunch of other things circled, etc. Presentation matters.

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